

hamishcoleman / thinkpad-ec Public

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 hamishcoleman Fix the build error and simplify the logic	799564b · last year
 .github/workflows	Oh, FFS, github actions, you cannot f... last year
 asm	CLEANUP: add old experimental x28... 7 years ago
 b580.H1ET73WW.img.d	Update the testing list so that we ca... 4 years ago
 b590.H9ET92WW.img.d	Update the testing list so that we ca... 4 years ago
 docs	Avoid confusion by using a more app... last year
 e330.H3EC35WW.img.d	Update the testing list so that we ca... 4 years ago
 l430.G3HT40WW.img.d	Update the testing list so that we ca... 4 years ago
 mec-tools @ 07a1b14	Update mec-tools to get a couple of f... 10 years ago
 nuvoton-tools	Import trivial checksum helper. last year
 radare	Fix x220 ec firmware image file in ra... 9 years ago
 scripts	Add partial support for p15g2 4 years ago
 t430.G1HT34WW.img.d	Add a rudimentary config file - this (f... 7 years ago
 t430.G1HT35WW.img.d	Add a rudimentary config file - this (f... 7 years ago
 t430.G1HT36WW.img.d	Port keyboard patch forward for t43... 7 years ago
 t430s.G7HT39WW.img.d	Add a rudimentary config file - this (f... 7 years ago
 w530.G4HT39WW.img.d	Add a rudimentary config file - this (f... 7 years ago
 x230.G2HT35WW.img.d	Add a rudimentary config file - this (f... 7 years ago
 x230t.GCHT25WW.img.d	Add a rudimentary config file - this (f... 7 years ago
 .gitignore	Ensure we ignore all the expected do... last year
 .gitmodules	Add x230 image and infrastructure 10 years ago
 .travis.yml	Strange, getting a permission error ... 7 years ago
 Descriptions.txt	Avoid confusion by using a more app... last year
 LICENSE	Add License details 10 years ago
 Makefile	Fix the build error and simplify the lo... last year
 README.md	Clarify that the stick should never be... 3 years ago
 autoexec.bat.template	Remove most of the variables from t... 9 years ago
 defconfig	Add a rudimentary config file - this (f... 7 years ago

t530.G4HT39WW.img.d

Add t530 support - this appears to s...

10 years ago

README GPL-2.0 license



COMPATIBILT Y WARNING:

As the result of CVE-2019-6171, newer Lenovo firmware update files have added a digital signature. If you upgrade to locked version you will not be able to patch your EC without downgrading it.

laptop	last modifiable	first protected version
t430	BIOS 2.81 (G1ETC1WW) EC 1.13 (G1HT35WW)	BIOS 2.82 (G1ETC2WW) EC 1.14 (G1HT36WW)
t430s	BIOS 2.75 (G7ETB5WW) EC 1.16 (G7HT39WW)	BIOS 2.76 (G7ETB6WW) EC 1.16 (G7HT40WW)
t530, t530i	BIOS 2.76 (G4ETB6WW) EC 1.13 (G4HT39WW)	BIOS 2.77 (G4ETB7WW) EC 1.14 (G4HT40WW)
w530	BIOS 2.75 (G5ETB5WW) EC 1.13 (G4HT39WW)	BIOS 2.76 (G5ETB6WW) EC 1.14 (G4HT40WW)
x230	BIOS 2.75 (G2ETB5WW) EC 1.14 (G2HT35WW)	BIOS 2.77 (G2ETB7WW) EC 1.15 (G2HT36WW)
x230t	BIOS 2.73 (GCETB3WW) EC 1.14 (GCHT25WW)	BIOS 2.75 (GCETB5WW) EC 1.15 (GCHT26WW)

Basically, any BIOS update package where the changelog mentions CVE-2019-6171 will have this lockdown.

Lenovo is tracking their response to this CVE at: <https://support.lenovo.com/gb/en/solutions/len-27764>

If you upgraded your BIOS to the locked version:

- Ensure that downgrading is possible in BIOS settings (Security/UEFI BIOS Update Option/Secure Rollback Prevention -> Disable)
- Downgrade it to the latest supported version. EC will be automatically downgraded as well

Intro

The main purpose of this software is to patch the EC on xx30 series thinkpads to make the classic 7-row keyboards work. There are also patches included (but disabled by default) to disable the authentic battery validation check.

With the patches included here, you can install the classic keyboard hardware on many xx30 series laptops and make almost every key work properly. The only keys that are not working are Fn+F3 (Battery) and Fn+F12 (Hibernate).

The xx30 keyboards do not have a Caps Lock Indicator and the motherboard has no hardware support for a Caps Lock Indicator, so the replacement classic keyboard will never turn on the Indicator on any laptop.

- A full writeup of the hardware modifications needed can be found at: http://www.thinkwiki.org/wiki/Install_Classic_Keyboard_on_xx30_Series_ThinkPads
- More information for hacking on this can be found in the [HACKING doc](#).
- A video presenting how these thinkpad laptops were hacked is online: <https://www.youtube.com/watch?v=Fzmm87oVQ6c>
- Reverse engineering details for the L430, L530, B580, V480, V480c, V580, V580c and E330 models can be found in the [thinkpad-Lx30-ec project](#). Patches for patching the genuine battery check on these have been contributed from that project. (These all have an Nuvotron EC, with a different architecture to the original "classic" thinkpad models).

Step-by-step instructions:

This software expects to be run under Linux (real Linux, not Microsoft Windows Linux subsystem). For best results, ensure you have updated your BIOS to a recent version before starting. If there is too large a difference between the BIOS and EC versions then the flash process will not complete.

A little more detail about the BIOS versions: It is not so much a question about upgrading to a recent BIOS version, but more of ensuring you are using a compatible EC firmware version. For safety, ensure that the EC version you are running is the same as the EC version used by the patched image you build. The version used to build the patch is shown at the end of the build process and during the pre-flash warning message.

1. Ensure you have installed the minimum required packages On Debian, this can be done with:

```
sudo apt-get update
sudo apt-get install make git
```



On Fedora, you could install it with dnf:

```
sudo dnf install git mtools openssl-devel
sudo dnf group install "C Development Tools and Libraries"
```



On OpenSUSE, try:

```
sudo zypper in git mtools libressl-devel
```



2. Clone a copy of this repo on to your computer:

```
cd ~/
git clone https://github.com/hamishcoleman/thinkpad-ec
```



3. Change to the directory created by the clone:

```
cd ~/thinkpad-ec
```



4. Install the prerequisite packages On Debian, this can be done with:

```
sudo make build-deps
```



5. Show the list of laptops and USB image file names:

```
make list_laptops
```



6. Choose your laptop model name from the list shown. E.G. "patched.x230.img" for a x230 laptop.

7. Optionally, the configuration can be changed from the defaults at this point. Read the [CONFIG doc](#) for details of the available config options.

8. Using the name chosen in the previous step, make the fully patched image for this laptop (this will download the original file from Lenovo and patch it):

```
make patched.x230.img
```



9. Insert your USB stick and determine what device name it has. (Note: chose a USB stick with nothing important on it, it will be erased in the next step) This command should help you find the right device:

```
lsblk -d -o NAME,SIZE,LABEL
```



Note: Do not mount the USB stick. If your desktop environment automatically mounts devices for you, you will need to unmount the stick. You may find that using the "Eject" option does not work as it may turn the power off to the stick, which will stop the next step from working.

10. Write the bootable patched image onto the USB stick device (replace the "sdx" in this command with the correct name for your usb stick)

WARNING: if you do not have the right device name, you might overwrite your hard drive!

```
sudo dd if=patched.x230.img of=/dev/sdx bs=4M status=progress conv=fsync
```



Your USB stick is now ready to boot and install the patched firmware.

Notes:

- You can also create a bootable CDROM image for burning to a disk by asking for a ".iso" file instead of the ".img" in step 6 above. Then you can use your normal CDROM burning tools to put this image on a blank cd and boot it up, skipping steps 7 and 8.
- The configuration is applied during the building of the patched image. If you wish to change the configuration, the patched image will need to be rebuilt.

Booting the stick and flashing the firmware:

While flashing the firmware is as simple as booting the USB stick created above, there are a couple of steps that can help the process. This is more a list of issues that the community has discovered as the patch was applied in different circumstances than a hard and fast set of requirements.

The flashing process takes place in two distinct steps (these are outlined below, but explained in more detail in [firmware_flashing doc](#))

1. Booting the USB stick:

- First shows a page with information about the patch, including which laptop type it was built for.
- Then it hands the new EC update to the BIOS, "staging" it for a future flashing into the EC hardware
- Finally it reboots the system.

2. Under the BIOS control, during a bootup:

- During the boot, the BIOS notices that it has a new EC update staged
- It then checks if it is safe to flash this update to the EC.
- If everything is safe, it will show a screen saying "Flashing EC"
- The system will bootup normally with the new EC code running.

If you don't see this second screen with the "Flashing EC" message, your EC has not been flashed, and you should continue reading below to see what steps you can take to ensure the EC is properly flashed with the patched firmware. In this cases everything might look like it was successful but after the reboot the keys are not remapped.

- For best results, ensure you have the power charger plugged in during the flashing process.
 - Some chargers seem to have issues with actually performing the flashing procedure after the flash process reboots. So, if you have - or can borrow - other chargers, try that.
-

The firmware flash process generally requires you to have a charged battery plugged in to the laptop before it will complete.

- It may be possible to bypass the requirement for a charged battery if you unplug the battery completely.
- Alternatively, it might be simply looking for any battery /and/ the power charger plugged in.

Yes, this is contradictory, but it is worth trying both options.

- An ultrabay battery is not considered by the update mechanism to be a suitable source of power - when trying different battery options, ensure you are trying batteries in the main battery slot.
- Ensure your BIOS has been configured to boot from "Legacy" and not "UEFI" before trying to boot.
- If you do normally use UEFI boot, there has been at least one case where the EC does not get flashed until

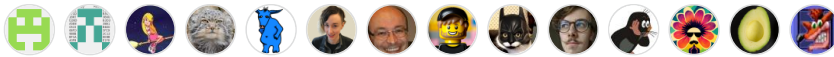
Releases

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